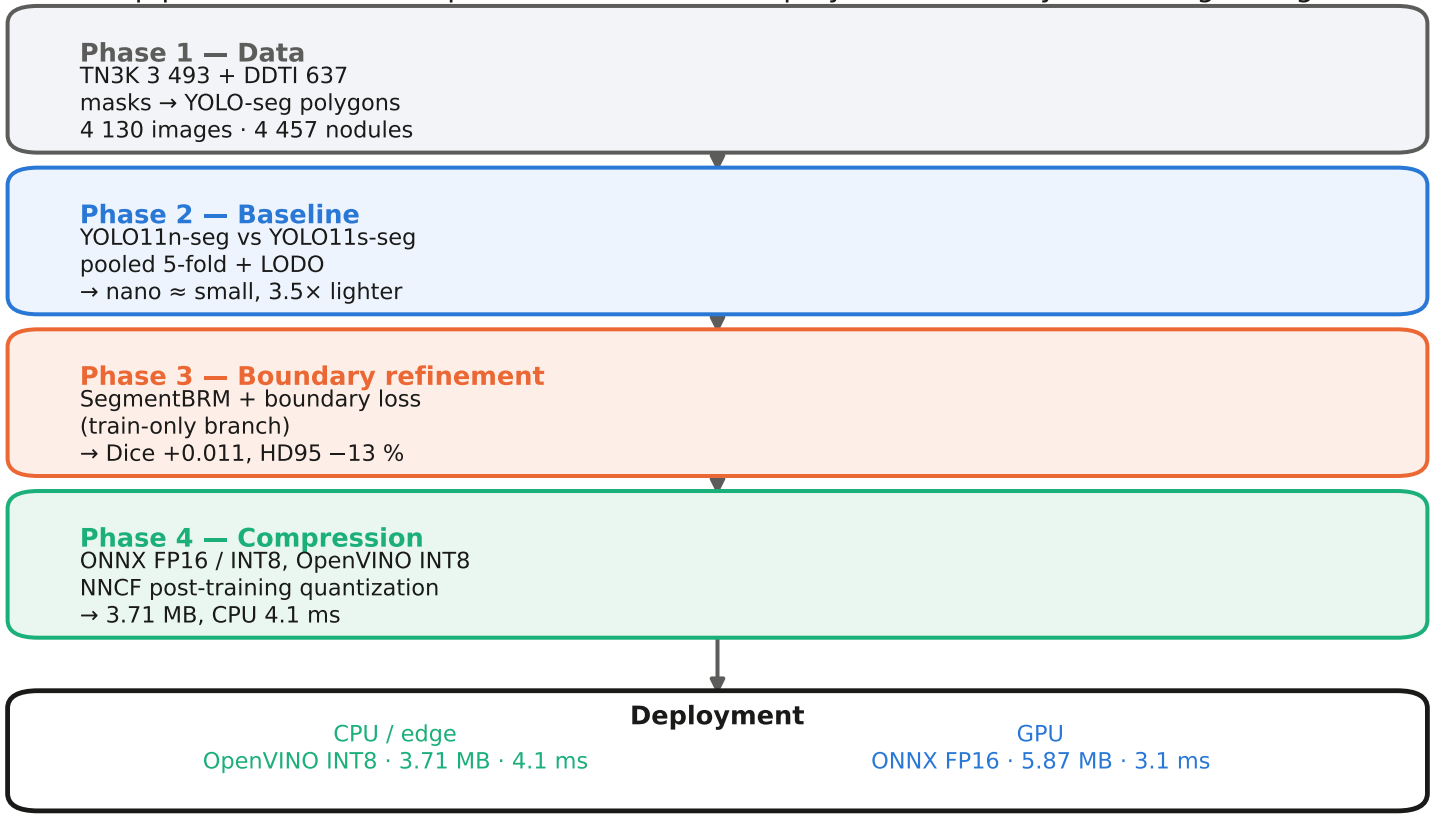Research pipeline: from mixed public datasets to a deployable boundary-refined lightweight model